WEAKLY ETALE MAPS ARE STABLE UNDER BASE CHANGE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The base change of a flat ring map is flat. invoked at proof lines 30316-30318. Gives part (1) and is reused for part (2).

Source reference. Original lemma 00HI: sources/stacks-chow/tags.txt, source lines 644–644. Cited at proof lines 30317-30318 for the flatness of the base-changed multiplication map; the conclusion is used inside the counted span.

Lemma 0.2. *Let $A \to B$ and $A \to A'$ be ring maps. Let $B' = B \otimes_A A'$ be the base change of B.*

(1) *If $B \otimes_A B \to B$ is flat, then $B' \otimes_{A'} B' \to B'$ is flat.*
(2) *If $A \to B$ is weakly étale, then $A' \to B'$ is weakly étale.*

Proof. Assume $B \otimes_A B \to B$ is flat. The ring map $B' \otimes_{A'} B' \to B'$ is the base change of $B \otimes_A B \to B$ by $A \to A'$. Hence it is flat by Algebra, Lemma 00HI. This proves (1). Part (2) follows from (1) and the fact (just used) that the base change of a flat ring map is flat. □

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